§5b — Stratification is Forced by an Impossibility (draft for integration)

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Stratification is forced by an impossibility, not chosen as a refinement

The impossibility. An anytime-valid e-process E_t certifies $\mathbb{E}[E_\tau] \le 1$ at every stopping time τ. That guarantee is *marginal*—averaged over the distribution of test points the monitor happens to see. It does not, and cannot distribution-free, certify the conditional statement $\mathbb{E}[E_\tau \mid \text{difficulty} = d] \le 1$ for every difficulty level d. This is structurally analogous to the conditional-coverage impossibility of Barber, Candès, Ramdas, and Tibshirani (arXiv:1903.04684)[1]: by an analogous argument, with no assumptions on the data-generating distribution and in finite samples, marginal and conditional validity cannot be had together. The gap is structural, not an artifact of a betting rule.

Why it bites the co-failure monitor specifically. One is tempted to route around difficulty analytically. The natural candidate is the detection drift

$$\mu \;=\; \mathrm{KL}(P_{\text{joint}} \,\|\, P_{\text{marg-product}}),$$

the Kullback–Leibler divergence between the observed joint failure law and the product of its marginals. Because $\mu = 0$ iff the two judges fail independently, μ is a pure dependence functional, and it is tempting to call it "difficulty-orthogonal by construction"—whatever the base failure rate, μ sees only the coupling. This is false. Difficulty re-enters the e-process through two channels that do not cancel—the variance of each log-likelihood increment scales with the marginal $p(1-p)$, and estimation error in the nuisance marginals propagates into the realized increments—so for Bernoulli failure indicators, mutual information is *not* a copula invariant, and difficulty cannot be conditioned out for free. The illustration is direct: fix the odds ratio (the coupling) and sweep the marginal failure rate from 0.05 to 0.5, and μ moves by an order of magnitude. The only sense in which μ "conditions out" difficulty is circular: define "same coupling" to mean "same μ", which defines the problem away rather than solving it.

Consequence. So the drift law, latency $\propto 1/\mu$, survives as the quantitative heart of the monitor—but not as a difficulty-robust one. Its non-robustness *is* the conditional-coverage gap: no distribution-free functional of the stream can deliver per-difficulty validity for free.

[1] That result is a conformal-prediction theorem, not an e-process result; we invoke it as a structural analogy only.

Stratification is what the impossibility leaves open—the one escape it does not close.[2] We stratify not to refine the estimator but because the theorem forbids the alternative.

The positive construction. Partition the difficulty axis into K strata and run a separate betting e-variable within each. On items in stratum k,

$$e \;=\; 1 + \lambda\big(U - V - \delta_k\big),$$

where $U = \mathbf{1}\{\text{both judges fail on item } i\}$ and $V = W_i^s W_j^t$ pairs the two judges across *different* items $s \neq t$ drawn from items other than the current item, $s, t \neq i$, so the marginals multiply exactly, $\mathbb{E}[V] = ab$, with no fitted nuisance. The slack $\delta_k = 2\varepsilon$ absorbs the worst-case two-sided excursion of the stratum's base rate, and the betting parameter is confined to $\lambda \in [0, 1/(1+2\varepsilon)]$ so the e-variable stays non-negative. The product over strata (or a mixture across k) is again a valid e-process by Ville. Conditional coverage is recovered by construction—one certificate per stratum—at the $O(K)$ cost the impossibility theorem says such coverage must carry.

The practical difference. On a synthetic panel where difficulty and coupling are confounded, the three methods behave as follows:

Method	Description	Type-I error
A	Stratified cross-item monitor	0.00
B	Unstratified cross-item monitor	0.23–0.44
C	Naïve plug-in (fitted marginals)	0.59–0.70

Stratification is not tightening a bound; it is the difference between a valid test and an invalid one. (Synthetic; a proof of necessity, not an empirical effect size.)

A second conditioning variable, and a distinction we keep. Difficulty is not the only axis on which a panel's ballots carry structure. Shu's pivotality analysis (arXiv:2608.06940) conditions on the vote margin $m_i = |2s_i - k|$: nearly all of a panel's recoverable accuracy sits on the $\approx 16\%$ [3] of queries where a single flip would change the verdict, and, unlike difficulty, margin needs no labels to compute. It is tempting to treat difficulty and margin as interchangeable stratification axes governed by one impossibility. They are governed by the same impossibility—neither can be conditioned out for free—but they are not interchangeable, and the difference is where our construction lives.

Difficulty is *ancillary* to the tested dependence: conditioning on it leaves the two judges' ballots conditionally independent, so $\mathbb{E}[V] = ab$ is preserved stratum by stratum. Margin is *not* ancillary; it is near-sufficient for the co-failure alternative, because co-failure *is* excess agreement and margin is a function of the agreement count. Conditioning on the ballot sum $s_i = c$ pins each judge's marginal to c/k and induces a negative within-item covariance,

$$\operatorname{Cov}(W_i, W_j \mid s_i = c) \;=\; -\frac{(c/k)(1 - c/k)}{k-1} \;<\; 0,$$

which breaks the exact factorization $\mathbb{E}[V] = ab$ that the whole margins-free construction rests on—and drains the test's power precisely at unanimity, where co-failure is strongest. The distinction is that difficulty is *exogenous* to the ballots—so we stratify on it—whereas margin is *endogenous*, a deterministic function of the very agreement count the test interrogates. We therefore keep margin strictly as a *post-hoc validity diagnostic*—a label-free check that Type-I control does not depend on where on the margin scale the panel sits—and *not* as a stratification

[2] The direct escape aside, shape constraints and DKW corrections may offer alternatives we have not yet closed.
[3] The $\approx 16\%$ figure is Shu's; pending our own primary read (arXiv:2608.06940).

variable co-equal with difficulty. This resolution is ours. Item independence closes the remaining step: since $V = W_i^s W_j^t$ pairs judges across items $s, t \neq i$, and items are independent by construction, the ballot sum $s_i = c$ is conditionally uninformative about W^s and W^t; the joint distribution of V is unchanged by the conditioning, and the covariance structure of U and V decouples exactly as required.

The perspectivist objection scopes the null, it does not refute it. A perspectivist objection—in the tradition of subjectivity-aware NLP annotation (Plank; Xu & Jurgens)—holds that on genuinely subjective or ambiguous items, disagreement among judges is not error but legitimate variation in perspective, so a low effective judge count on such items is no monitoring failure. We grant this, and it sharpens the null's scope rather than threatening the monitor's validity. The co-failure monitor targets items with a *determinate* ground truth, where correlated judge failure is a genuine reliability defect; on genuinely-indeterminate items an effective count of ≈ 2 is not a pathology to flag but real perspective spread, and the monitor should stay silent. The remedy is exactly the move we made for difficulty—stratify by ground-truth-determinability—because determinability is *exogenous* to the ballots (a property of the item, like difficulty), hence a legitimate stratification variable. We concede that determinability, like difficulty, is not directly observed and must be supplied by external (meta-)annotation, which places it on the same exogenous footing rather than making it freely readable off the stream. Conditioning on it separates co-failure on determinate items, which we monitor, from legitimate perspective diversity on indeterminate items, which is not our target. The perspectivist critique thus scopes the null; it does not refute it.

Alternatives under investigation (brief). Two alternatives to explicit stratification are under investigation and not yet adopted: shape-constrained e-processes (Wang–Ramdas, arXiv:2604.20612), which would monitor whether the failure-rate curve keeps its monotone/unimodal shape in difficulty rather than stratifying it out; and DKW-corrected conditional test martingales (arXiv:2602.13848). Both are flagged pending primary reading and are mentioned only for completeness.